

A. Cross-city model training

35 training cities + 5 validation cities

source-city positive OD intensities



Cross-city inputs

tract features + spatial graph
+ pairwise distances



Train and select model M_0

cross-city learning only



Freeze all model parameters

before target-city inference

**No target-city OD intensity
is used for model training**

B. Held-out target-city inference

Held-out target city

features + distances + known
positive support Ω_c



Frozen M_0

no fine-tuning or parameter update



Baseline prediction $T^{(0)}$

Target positive ground-truth OD flows

same target reference flows used for evaluation



Deterministic distance-bin aggregation

$Y_{D,c}$ oracle

C. Inference-time calibration

$T^{(0)} + Y_{D,c}$ oracle

on the same known positive support Ω_c



Compare bin shares

Aggregate predicted flow by distance bin
Compare baseline and target bin shares



Rescale flows within each bin

Multiply by ratio of target to baseline share



Calibrated prediction $T^{(1)}$

evaluate on the same Ω_c

Frozen model: no parameter update

Same support: no new OD links

Preserve support, total flow, and within-bin ranking

Dashed path: oracle intervention, not external telemetry